Trigonometric functions on the unit circle

Core Concepts: Trigonometric functions on the unit circle

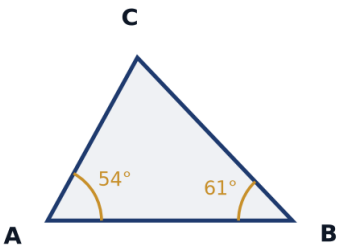
1 For the model $f(x) = 4x + 4$ in trigonometric functions on the unit circle, evaluate $f(2)$.

a. 12

b. 16

c. 8

d. 16



2 Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

3 A data set recorded for trigonometric functions on the unit circle consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 4x + 4$ in trigonometric functions on the unit circle, evaluate $f(3)$.

a. 16

b. 20

c. 12

d. 20

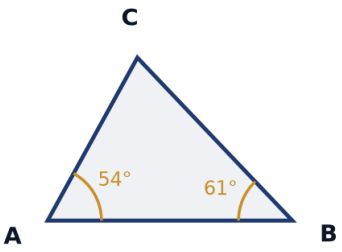
5 Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$



6 A data set recorded for trigonometric functions on the unit circle consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

Graphical & Geometric Interpretation

7 For the model $f(x) = 4x + 4$ in trigonometric functions on the unit circle, evaluate $f(4)$.

a. 20

b. 24

c. 16

d. 24

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8 Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

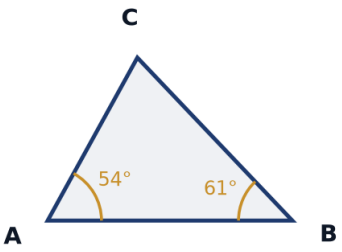
9 A data set recorded for trigonometric functions on the unit circle consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5



Exam-Style Problem Solving & Modeling

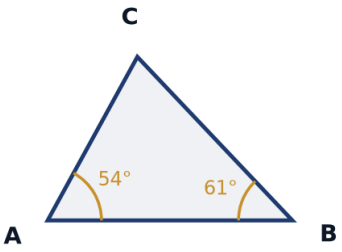
10 For the model $f(x) = 4x + 4$ in trigonometric functions on the unit circle, evaluate $f(5)$.

a. 24

b. 28

c. 20

d. 28



11 Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

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12 A data set recorded for trigonometric functions on the unit circle consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

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